

TP Intro Supervised Answers

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Theoretical questions

OLS

$\beta^* = (x^\top x)^{-1} x^\top y = Hy$. Let $\tilde{\beta} = Cy$ another linear unbiased estimator of β s.t. $C = H + D$

Show that OLS is the estimator with the smallest variable

$$\mathbb{E}(\tilde{\beta}) = \mathbb{E}((D + H)y) = (D + H)x\beta = \beta^* + Dx\beta$$

$\tilde{\beta}$ is unbiased so : $Dx\beta = 0$, for every β so : $Dx = 0$.

$$\text{So : } \tilde{\beta} - \mathbb{E}(\tilde{\beta}) = \beta - \beta^* + D(y - x\beta) = \beta^* - \beta + D\epsilon$$

So we deduce that :

$$\text{Var}(\tilde{\beta}) = \mathbb{E}((\tilde{\beta} - \mathbb{E}(\tilde{\beta}))(\tilde{\beta} - \mathbb{E}(\tilde{\beta}))^\top) = \mathbb{E}((\beta - \beta^* + D\epsilon)(\beta - \beta^* + D\epsilon)^\top)$$

$$(\beta^* - \beta + D\epsilon)(\beta^* - \beta + D\epsilon)^\top = (\beta - \beta^*)(\beta - \beta^*)^\top + (\beta - \beta^*)\epsilon^\top D^\top + D\epsilon(\beta - \beta^*)^\top + D\epsilon\epsilon^\top D^\top$$

So, assuming that : $\text{Var}(\epsilon) = \sigma^2 I$, we deduce that :

$$\text{Var}(\tilde{\beta}) = \text{Var}(\beta^*) + \mathbb{E}((\beta - \beta^*)\epsilon^\top)D^\top + D\mathbb{E}(\epsilon(\beta - \beta^*)^\top) + \sigma^2 DD^\top$$

$$\text{We have : } (\beta - \beta^*)\epsilon^\top = (Hx\beta - \beta + H\epsilon)\epsilon^\top = (Hx\beta - \beta)\epsilon^\top + H\epsilon\epsilon^\top$$

$$\text{So : } \mathbb{E}((\beta - \beta^*)\epsilon^\top) = \sigma^2 H$$

$$\text{So : } \text{Var}(\tilde{\beta}) = \text{Var}(\beta^*) + \sigma^2 DD^\top + \sigma^2(HD^\top + DH^\top)$$

$$\text{We have : } HD^\top = (x^\top x)^{-1} x^\top D^\top = (x^\top x)^{-1} (Dx)^\top = 0$$

$$\text{and } DH^\top = Dx(x^\top x)^{-1} = 0$$

$$\text{SO : } \text{Var}(\tilde{\beta}) = \text{Var}(\beta^*) + \sigma^2 DD^\top$$

$$\text{Var}(\beta^*) < \text{Var}(\tilde{\beta}) \Leftrightarrow DD^\top \succeq 0 \text{ (always true)}$$

The assumption of OLS we need to use is : Homoscedasticity : $\text{Var}(\epsilon) = \sigma^2 I$

Ridge Regression

- **Show that the estimator of ridge regression is biased.**

$$\text{Let } f(\beta) = (y_c - x_c\beta)^\top (y_c - x_c\beta) + \lambda \|\beta\|_2^2 = \|y_c - x_c\beta\|_2^2 + \lambda \|\beta\|_2^2$$

$$\nabla_\beta f(\beta) = 0 \Leftrightarrow -x_c^\top (y_c - x_c\beta) + \lambda\beta = 0 \Leftrightarrow (x_c^\top x_c + \lambda I)\beta = y_c \Leftrightarrow \beta_{ridge}^* = (x_c^\top x_c + \lambda I)^{-1} x_c^\top y_c$$

We have :

$$\mathbb{E}(\beta_{ridge}^*) = (x_c^\top x_c + \lambda I)^{-1} x_c^\top \mathbb{E}(y_c) = (x_c^\top x_c + \lambda I)^{-1} x_c^\top x_c \beta \neq \beta \text{ for } \lambda \neq 0$$

So the Ridge estimator β_{ridge}^* is biased for $\lambda \neq 0$.

- **Recall that the SVD decomposition is $x_c = DV^\top$. Write down by hand the solution β_{ridge}^***

ridge using the SVD decomposition. When is it useful using this decomposition ?

Hint: do you need to invert a matrix ?

SVD decomposition of x_c is : $x_c = DV^\top$ where D is diagonal and V are orthogonal matrixes.

$$\beta_{ridge}^* = (VD^\top DV^\top + \lambda I)^{-1} VD^\top y_c = (VD^2 V^\top + \lambda I)^{-1} VD^\top y_c$$

$$\text{We have : } VD^2 V^\top + \lambda I = VD^2 V^\top + \lambda VV^\top = V(D^2 + \lambda I)V^\top$$

So :

$$(VD^2 V^\top + \lambda I)^{-1} = (V^\top)^{-1}(D^2 + \lambda I)^{-1}V^{-1} = V(D^2 + \lambda I)^{-1}V^\top$$

So we conclude that :

$$\beta_{ridge}^* = V(D^2 + \lambda I)^{-1}D^\top y_c$$

In this decomposition, the inverse of the matrix $D^2 - \lambda I$ is easy to compute since it is diagonal. (We only need to inverse the diagonal elements).

This method is computational efficient, stable and it is easy to make the tuning of λ , since the SVD parameters are pre-determined and independent of λ . So we only need to make the final computation to get the ridge estimator corresponding to that value of λ .

- **Remember that $Var(\beta_*) = \sigma^2(x^\top x)^{-1}$. Show that $Var(\beta_*) \succeq Var(\beta_{idge}^*)$**

We have :

$$Var(\beta_{idge}^*) = (x_c^\top x_c + \lambda I)^{-1} x_c^\top Var(\epsilon) x_c (x_c^\top x_c + \lambda I)^{-1} = \sigma^2 (x_c^\top x_c + \lambda I)^{-1} x_c^\top x_c (x_c^\top x_c + \lambda I)^{-1}$$

$$Var(\beta_{idge}^*) = V(D^2 + \lambda I)^{-1} D^\top \sigma^2 I D^\top (D^2 + \lambda I)^{-1} V^\top = \sigma^2 V(D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1} V^\top$$

$$Var(\beta_*) = \sigma^2 (VD^\top DV^\top)^{-1} = \sigma^2 (VD^2 V^\top)^{-1} = \sigma^2 V(D^2)^{-1} V^\top$$

$$Var(\beta_{idge}^*) = \sigma^2 V(D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1} V^\top Var(\beta_*) = \sigma^2 V(D^2)^{-1} V^\top$$

$$(D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1} = \left(d_i^2 (d_i^2 + \lambda)^2 \right)_{1id}$$

$$(D^2)^{-1} = \left(1d_i^2 \right)_{1id}$$

$$\text{We have, for } 1 \leq i \leq d : d_i^2 (d_i^2 + \lambda)^2 \geq d_i^2 (d_i^2)^2 = 1d_i^2 (\lambda = 0)$$

$$\text{So : } (D^2)^{-1} \succeq (D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1}$$

V is an orthogonal matrix, so we conclude that :

$$Var(\beta_*) \succeq Var(\beta_{idge}^*)$$

- **When λ increases what happens to the bias and to the variance ?**

Let's compute the MSE :

$$\mathbb{E} (y_0 - x_0^\top \beta_{idge}^*)^2 = \mathbb{E} (x_0^\top (\beta - \beta_{idge}^*) + \epsilon_0)^2 = \mathbb{E} \left(x_0^\top (\beta - \beta_{idge}^*) \right)^2 + 2 \mathbb{E} \left(x_0^\top (\beta - \beta_{idge}^*) \epsilon_0 \right) + \mathbb{E} (\epsilon_0^2)$$

$\epsilon_0 \perp \beta_{idge}^*$ so :

$$= \mathbb{E} \left(x_0^\top (\beta - \beta_{idge}^*) \right)^2 + 2 \mathbb{E} \left(x_0^\top (\beta - \beta_{idge}^*) \right) \mathbb{E} (\epsilon_0) + \sigma^2$$

$$= x_0^\top Var(\beta_{idge}^*) x_0 - \left(\mathbb{E} \left(x_0^\top (\beta - \beta_{idge}^*) \right) \right)^2 + \sigma^2$$

When λ increases, the coefficients of the model are shrunk towards 0.

$$Var(\beta_{idge}^*) = \sigma^2 V(D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1} V^\top$$

$$\text{with : } (D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1} = \left(d_i^2 (d_i^2 + \lambda)^{-2} \right)_{1id}$$

if λ increases, the diagonal elements of $(D^2 + \lambda I)^{-1} D^2 (D^2 + \lambda I)^{-1}$ decreases, and since V is orthogonal we deduce that $Var(\beta_{idge}^*)$ decrease.

$$\mathbb{E}(\beta_{ridge}^*) = (x_c^\top x_c + \lambda I)^{-1} x_c^\top x_c \beta$$

$$(x_c^\top x_c + \lambda I)^{-1} x_c^\top x_c \beta = V(D^2 + \lambda I)^{-1} V^\top V D^\top D V^\top \beta = V(D^2 + \lambda I)^{-1} D^2 V^\top \beta$$

So the bias :

$$= \mathbb{E}(\beta_{ridge}^*) - \beta = V((D^2 + \lambda I)^{-1} D^2 - I) V^\top \beta$$

As λ increases, the diagonal elements of $(D^2 + \lambda I)^{-1} D^2$ decrease, so the whole bias becomes farther away from 0.

- Show that $\beta_{idge}^* = \beta_* 1 + \lambda$ when $x_c^\top x_c = I_d$

We have : $x_c^\top x_c = I_d$ so :

$$\beta_{idge}^* = ((1 + \lambda)I)^{-1} x_c^\top y = 11 + \lambda x_c^\top y$$

$$\beta_* = x_c^\top y$$

So :

$$\beta_{idge}^* = \beta_* 1 + \lambda$$

Elastic Net

- Let's show that when $x_c^\top x_c = I_d$ $\beta_e^* = (\beta)_1 21 + \lambda_2$

We have $x_c^\top x_c = I_d$ so : $\beta_* = x_c^\top \beta$

$$(y_c - x_c \beta)^\top (y_c - x_c \beta) = y_c^\top y_c - 2\beta^\top x_c^\top y_c + \beta^\top x_c^\top x_c \beta = y_c^\top y_c - 2\beta^\top x_c^\top y_c + \beta^\top \beta$$

So we can deduce that :

$$(y_c - x_c \beta)^\top (y_c - x_c \beta) = y_c^\top y_c - 2\beta^\top \beta_* + \beta^\top \beta$$

So we deduce that the function to minimize becomes :

$$_0(\beta) = \underset{\beta}{y_c^\top y_c - 2\beta^\top \beta_* + \sum_{i=1}^d \beta_2 + \lambda_2 \sum_{i=1}^d \beta_2 + \lambda_1 \sum_{i=1}^d |\beta|}$$

So the final function to minimize is :

$$(\beta) = -2\beta^\top \beta_* + \sum_{i=1}^d \beta_2 + \lambda_2 \sum_{i=1}^d \beta_2 + \lambda_1 \sum_{i=1}^d |\beta|$$

Let 1 d , we have in terms of β :

$$(\beta) = -2\beta \beta_* + (1 + \lambda_2) \beta_2 + \lambda_1 |\beta|$$

$$(\beta) \beta = -2\beta_* + 2(1 + \lambda_2) \beta + \lambda_1 ig(\beta)$$

$$(\beta) \beta = 0 \Leftrightarrow \beta = 2\beta_* - \lambda_1 ig(\beta) 2(1 + \lambda_2)$$

So :

$$\beta_e^* = \beta_* - \lambda_1 2ig(\beta)1 + \lambda_2$$